GCSE

Rewarding Learning

Chief Examiner's and Principal Moderator's Report

Digital Technology



Summer Series 2019

Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2019 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's section on our website at www.ccea.org.uk.

Contents

Assessment Unit 1	Digital Technology	3
Assessment Unit 2	Digital Authoring Concepts	5
Assessment Unit 3	Digital Authoring Practice	11
Assessment Unit 4	Digital Development Concepts	14
Assessment Unit 5	Digital Development Practice	18
Contact details		20

GCSE DIGITAL TECHNOLOGY

Chief Examiner's Report

Subject Overview

This is the first year of the full award for both Digital Technology (Multimedia) and Digital Technology (Programming). It was pleasing to see candidates perform well across the range of abilities and providing a very good level of technical responses. The technical language and programming concepts addressed in both strands were well attempted throughout. Teachers are to be congratulated for embracing the new specifications and preparing candidates so well to undertake the examinations.

Assessment Unit 1 Digital Technology

Overview

Overall the paper was well designed and permitted candidates to achieve good outcomes, it also catered for all levels of ability. Some outcomes were perhaps lower than expected and this may be due to some candidates taking the paper in their fourth year at secondary school and their knowledge maybe somewhat limited and not used to external exams. The language within the paper was suitable for GCSE and the mark scheme provided enabled the markers to positively award candidates.

Question 1

MULTIPLE CHOICE

This question was made up of Parts (a) to (h), each part contained four short statements and the candidate was asked to choose the correct response.

Q1 **(a)** Well answered.

(b) Well answered but for many D was a common wrong answer.

(c) Well answered but C & D were common wrong answers.

(d) Well answered.

(e) Well answered.

(f) Well answered.

(g) Well answered but with A as popular wrong answer.

(h) Well answered.

Q2 This question was generally fairly well answered by the candidates, most were able to get one mark on Part (a)(i) but some failed to include an example. However, in relation to Part (a)(ii) many candidates failed to give an appropriate example that matched the definition.

(b) Was well answered.

Q3 Question three was a database question. Parts 3(a,b,e,f,) were very well answered. However, many candidates seemed to read Part 3(c)(i) incorrectly and didn't give a response about MEMBERID. Part 3(c)ii seem to cause some difficulty with the candidates, many failed to note that this field was connected or referenced to the DATEOFBIRTH field.

Q4 Question 4 was well structured, and the majority of candidates scored highly on this question with the exception of Part 4(a)(iii) where many did not know the dpi stood for Dots Per Inch.

Q5 This question was generally well answered but some candidates were unable to discern the difference between defragmenting and task scheduling. This may indicate that the more difficult aspect of theory around system software are challenging to many candidates. Part 5(b) is a question that often discriminates between the better prepared candidates and is included as a stretch and challenge question.

Q6 Question 6 was well attempted by all candidates. Part 6(a) didn't seem to pose much difficulty to the majority of candidates. In Part 6(b) the diagram worked well and the only aspect of that seemed to cause candidates difficulty was knowing the purpose of a terminator in a Bus Network. The use of the diagram in Part 6(d) was helpful in enabling candidates to answer the question and most candidates did not have any issue identifying the three parts of a Star Network. Some candidates struggle to be able to identify advantages and disadvantages generally and perhaps more attention needs to be given to having clear identifiable advantages and disadvantages when practicing such questions.

Q7 Question 7 is a quality of written communication question. This question is designed to permit the examiner to assess the ability of the candidate to write coherently and to be able to discuss such topics. This question required the candidate to describe the benefits that cloud computing has brought to gaming customers. This question was in the main well attempted and many candidates were able clearly to describes sensible benefits that the cloud has brought to the gaming fraternity. Many were able to write about half a page with some more able candidates writing three quarters of a page. Again such a question can be omitted by weaker candidates who do not like having to write detailed responses.

Q8 Question 8 was based on spreadsheets and was in the same format that the spreadsheet question was in the old GCSE ICT paper. While the easier parts of the question were well attempted it was somewhat disappointing that many candidates did not have the ability to read the graph and could not answer what an absolute cell reference is. This question was designed to determine that candidates can look at a graph which spreadsheets are generally designed to produce and to then be able to explain what data the graph is based on. Absolute cell referencing has always appeared on this type of question and it would be helpful for candidates to learn a simple definition of something they probably can carry out without much difficulty.

Part 8(f) appeared to be also a challenge to many candidates with very few being able to give a good example based on cell F1.

Q9 Question 9 like Question 7 requires the candidate to write in a more extensive way, this question was well attempted but numerous candidate's knowledge was out of date and many mixed the idea of teleworking into the question. Again, some confusion around understanding that while the company may not be able to afford to bring in trainers to do training sessions, there are still associated costs to the company and generally staff will be paid for the time they spend online training. This topic may require candidates to have a better understanding if there are future similar questions.

Assessment Unit 2 Digital Authoring Concepts

Overview

In this unit, candidates develop understanding of the concepts involved in the development of digital systems

Q1 Multiple choice questions were well attempted by the majority of candidates and a significant number of candidates achieved full marks in this questions

(a) The majority of candidates were correctly able to identify a data dictionary as being a table listing the field headings used in a database.

(b) Few candidates had difficulty recognizing the most suitable file formats for distributing video files across the internet.

(c) A small number of candidates incorrectly answered this question and for some the term 'merge field' was misconstrued to mean the data items stored in a merged word processed document.

(d) Most candidates were able to correctly identify option B as being the statement which correctly defined the term 'entity'.

(e) The term macro was easily described by the majority of candidates responding to this question part.

(f) Most candidates were correctly able to select the correct definition of the term template.

(g) This question posed some difficulty for candidates with a significant number confusing the and response options.

(h) It was pleasing to see a large proportion of candidates correctly choosing option D as the correct response to this question.

(i) With relational databases playing a large part in the assessment of this unit it was pleasing to see most candidates were able to correctly select Entity Relationship diagram as the correct response to this part of Question 1.

Q2 This was a question on spreadsheets worth a total of 10 marks.

(a) Candidates were asked initially to use an example from the context provided to explain the term data redundancy. While most candidates were correctly able to provide a definition of the term, a large number did not select an appropriate answer from the context provided and therefore were not able to access the second mark in this question part. Where candidates were not awarded a mark for the definition they provided this was in most cases due to the fact that they failed to mention 'unnecessary' when referring to duplication of data.

(b) **(i)** It was pleasing to see a large proportion of candidates correctly reference the accuracy or consistency of data as being a term determining factor in its integrity; correctness was also a common (and accepted) response.

Most candidates were also correctly able to recognise suitable examples of poor data integrity with reference to Jamie Duddy/James Duddy being the most commonly provided example. Some candidates were not awarded marks for their response to this question part as their responses were too generic, for example 'names are repeated'.

(c) Most candidates correctly selected the third option as providing the correct criteria required to produce the appointments specificed. Where candidates were not awarded the mark for this question part, it was in the most part due to confusion over the greater than and the less than symbols presented in the criteria; in these cases the candidates incorrectly selected the first option on the table provided.

(d) Most candidates correctly expanded SQL but a common incorrect response was Select Query Language.

(e) Candidates who correctly referenced the TBLAppointments example provided at the start of this question were awarded full marks in this question part. Other candidates who did not refer to the table provided were unable to use the correct field names in the response; instead these candidates provided their own interpretation of the field names they would expect to see in a SQL statement of this type and generic terms such as TABLE and NAME were provided as answers. Teachers should remind candidates of the importance of reviewing prompts and supporting graphics when answering context based questions.

(f) It was pleasing to see that most candidates understood the concept of calculated fields in database reports. However a large number of candidates could not be awarded the marks available to this question part as their responses did not correctly reference the field headings provided in the report structure. The examining team were happy to accept a description of the formula required when calculating Grand Total but candidates were expected to reference the correct fields in the report for first mark and a reference to the '+' sign or the term add/addition for the second mark.

Q3 **(a)** This question part was well answered in the main.

It was pleasing to see the majority of candidates were able to access at least 50% of the marks for this question part. Candidates should however be encouraged to provide elaboration on responses to questions were more than one mark is being provided for a candidate response as this will help ensure all marks are accessible to the candidates.

Popular responses included reference to the early detection of errors and subsequent application of required modifications, the fact users needs are more likely to be met due to the fact that misunderstanding between the user and developer were reduced.

(b) Most candidates were able to correctly identify that end users were involved at all stages during this model of development. However fewer candidates included the necessary elaboration which referenced the provision of feedback to ensure that both marks could be awarded for their response to this question part.

Q4 This question focused on testing and examined candidates understanding of system testing, definitions of various methods of testing and the development of appropriate test plans.

(a) The examining team was pleased to see that most candidates were correctly able to provide definitions of both black box and white box testing. For many candidates the responses contained reference to the fact that in black box testing the design or structure of the underlying solution was unknown by the tester whilst in white box testing the tester was familiar with the code. Other common responses included reference to the fact that white box testing was carried out by a programmer. In future examination of this topic area the examining team would expect to see an increase in the technical responses provided by candidates. For example: -

Black box – focuses on inputs/outputs, tests functionality, relates to how the interface operates

White box – employs the use of trace tables, tested using a dry run process

(b) **(i)** In this question part, candidates were expected to complete a test plan by entering appropriate test data for three test items. Most candidates correctly interpreted the context of the question and therefore correctly attempted the question. Marks allocated were however varied with some candidates incorrectly describing the test data they would employ in the testing process. For example, some candidates opted to leave the third response option empty when they should have responded with a description of the fact that the database field should be left blank or the enter key would be pressed.

(ii) Most candidates were correctly able to select the second statement as being correct within the context of system testing.

(c) **(i)** Most candidates were correctly able to select the correct two statements which were true in relation to alpha testing.

(ii) While a large number of candidates were able to correctly reference the fact that A/B testing required the use of two versions of the same software solution, very few candidates in this paper were able to access the second mark in this question part by indicating that a comparison process was then employed to determine which was the most successful of the two software versions.

Q5 **(a)** **(i)** Most candidates were successfully awarded some marks for this question part which focused on effective testing of multimedia solutions. Candidates in the main failed to reference two key points when providing their response to this question part. Where more than one mark is awarded for a response, candidates must be encouraged to elaborate on key points. Common answers provided by candidates included mention of the need to check the images loaded but few went on to give consideration to the need to test that the images loaded/played automatically, that users can navigate back and forward through the images at their own pace for example.

(ii) This question asked candidates to consider how links to social media sites could be tested. Too many candidates provided generic responses containing terms such as ‘click on it’, ‘check it takes you to the correct place’. Expected responses included reference to the need to ‘click on the link’ to ensure it ‘took users to the correct website’.

(b) This question was well answered. Most candidates were awarded a mark for correctly identifying the fact that the company logo did not appear correctly due to the fact that the image was in an incorrect format/file type not supported, the image was deleted or the image was in the incorrect folder.

(c) Candidates remain unfamiliar with the main methods associated with incorporating scripted elements into html files despite this being specifically referenced in the unit 2 specification.

(d) The real world nature of this question made it accessible to most candidates. Almost all candidates were correctly able to identify two additional interactive features provided by online shopping applications.

Q6 Candidates demonstrated a good understanding of the various approaches taken to providing methods of interaction between users and digital applications.

(a) Most candidates fully understood and were able to describe features of a GUII which made it suitable for both inexperienced and experienced users.

(b) While most candidates understood that users could input instructions either manually or verbally using a natural language interface, too many candidates offered one word responses to this question, such as speak or type. One word answers are not acceptable at this level and schools should encourage more elaboration/description when responding to questions of this type.

(c) This question was well received by candidates who were able to draw upon their own experiences of using natural language interfaces when interacting with telephones. Most candidates were able to score at least two out of the four available marks for this question part.

(d) When referring to disadvantages of touch screen technologies, a significant number of candidates made reference to the fact that if wet or cold fingers would not work properly when inputting data. Candidates should be reminded that responses should focus on the technology rather than the user in questions of this type.

(e) This question was well answered with most candidates being awarded full marks. Closed response questions of this type are very successful in assessing candidate knowledge of some of the more technical elements of this specification.

(f) Most candidate responses to this question on the use of motion tracking technology focused correctly on the fact that the technology helped make the gaming experience more realistic, that it supported use by individuals with limited mobility or users with limited technological experience. This question was well answered.

(g) This question was not well answered with many candidates making reference to points such as the ability to increase text size or text colour when really what they should have been referencing was the ability to change screen colour settings or provide onscreen magnifiers for example.

(h) This was the first QWC question in this written examination. It was worth six marks and candidates were provided with hints relating to the structure of their response by being encouraged to give consideration to plug-in software and PDF documents. The majority of candidates were able to correctly provide a detailed description of at least one of the compatibility aids. Few candidates were able to correctly elaborate on both. Candidates should be well versed on the demands of QWC questions before entering their final examination. A full understanding of the need to incorporate both technical detail and technical terminology into extended answers is needed if candidates are to access level 3 in their responses to these question types. In this instance too many candidates relied on their ability to describe both compatibility aids rather than being able to expand answers to incorporate how they support cross-platform compatibility; this approach meant that the majority of candidates could not be awarded marks beyond level two in this instance.

Q7 Candidates respond well to questions which draw upon their real-life experiences, this question afforded many candidates the opportunity to access full marks in most question parts.

(a) **(i)** While most candidates were familiar with the term feedback and the types of feedback available to users in gaming applications some candidates missed the opportunity to achieve full marks by not referencing distinct examples of feedback as part of their response. Reference to items such as health bars/scores/progress bars are too similar and will only be credited once. Schools should encourage candidates to provide a broad range of examples in response to questions of this type.

(ii) This question part was well answered, as expected. The majority of candidates made reference to the fact that interactive elements made the game more realistic for the end user and that it made the gaming experience more enjoyable. As with Part (i) however some candidates missed the opportunity to reach full mark capacity with this question part by providing answers which were too similar, for example by referencing the immersive nature of the application alongside the fact that it makes the gaming experience more enjoyable.

(iii) This question was particularly well answered by candidates with most being awarded full marks for their response. Candidates were very aware of the impact of including interactive and multimedia elements on computer games.

(b) Almost all candidates were correctly able to define the term target audience.

Q8 Candidates own experiences with social media applications ensured this question was accessible and well answered with most candidates being awarded full marks.

(a) For many candidates the ability to post content or like other people posts proved to be the two features which made social media applications enjoyable to use.

(b) Concerns over location tracking, abuse and lack of socialisation proved to be the most problematic for the majority of users.

Q9 This question proved difficult for some candidates.

(a) Despite the fact that candidates create hyperlinks during the completion of their controlled assessment tasks, few candidates were able appropriately explain how hypertext is used in web creation. Candidates who were awarded marks for their response to this question were often only awarded one of the two available marks as most failed to reference the fact that it was used to create a hyperlink which allowed users to move from one page to another.

(b) Most candidates correctly recognized that the use of folders helps ensure that resources can easily be located when creating website applications.

(c) This question was well answered by most candidates.

Q10 This question focused on testing and examined candidates understanding of system testing, definitions of various methods of testing and the development of appropriate test plans.

Most candidates were correctly able to provide definitions of both black box and white box testing. For many candidates the responses contained reference to the fact that in black box testing the design or structure of the underlying solution was unknown by the tester whilst in white box testing the tester was familiar with the code.

Other common responses included reference to the fact that white box testing was carried out by a programmer or someone who have detailed knowledge of the code. In future examination of this topic area the examining team would expect to see an increase in the technical responses provided by candidates. For example: -

Black box – focuses on inputs/outputs, tests functionality, relates to how the interface operates

White box – employs the user of trace tables, tested using a dry run process

(b) Many candidates were able to select the correct statements about system and integration testing obtaining two marks.

Q11 The focus on this question was the process of evaluation and use of a range of document elements in support of the evaluation process.

(a) While many candidates were able to reference the fact that the main aim of the evaluation process was to gain feedback from the end user, very few were awarded the second mark for this question. Schools should ensure candidates are aware of the need to conduct an evaluation to ensure that all user requirements have been met or to help improve the product being developed

(b) It was pleasing to see the majority of candidates were able to correctly reference two individuals involved in the evaluation process; reference to the developer/ programmer and the end user were the most common responses provided

(c) This question was poorly attempted. While candidates are familiar with the need and content of each of the document types referenced in each part of this question it is important also that they are aware how various document types are used to support the evaluation process.

(i) Only a small number of responses provided made reference to the use of the user requirements document as a check list or point of comparison; too many candidates instead referenced the content of the document; describing what the document contained rather than how it was used.

(ii) This question part was the least well answered part of this question. Common responses from candidates focused on the testing process rather than how test plans were used during evaluation, for example, to see if…., to make sure everything works. Only a small percentage of candidates were awarded full marks to this question part.

(iii) The focus of most answers in this question part referenced how the design document contained plans for the proposed solution, rather than referencing the fact that these plans were used as a point of comparison either at the design stage or at the end of development.

Q12 This question asked candidates to match terms to their definition and was well received. Many candidates achieved at least 5 of the 7 marks available. The definitions for pointer, window, compression, icon and animation were most commonly correct, whilst the definitions for parameter and foreign key were incorrect.

Principal Moderator's Report

Assessment Unit 3 Digital Authoring Practice

Overview

All centres successfully submitted controlled assessment tasks electronically. Centres **must** ensure candidates have been provided with the correct scenario for Units 3. This information is clearly provided on the front cover of the controlled assessment units. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised. The quality and assessment of candidate work has remained accurate.

Administration

The following procedures must be adhered to:

- Samples should be submitted on **memory pen (preferably)**, or **CD**, rather than DVD.
- Unit 3 samples must be submitted **on separate media.**
- **The contents of each sample must be verified** after the sample has been copied to media.
- All candidates of the requested sample, including the **top** and **(complete) bottom candidate** of the sample must be included. A number of centres did not include the top and bottom candidate this series.
- **Folder labels 'top'** and **'bottom'** must be included after the 'candidate number' label on the sample folder, for example, 'Candidate 0000 (Top)'.
- Only **essential files** should be submitted for moderation. Original data files or duplicate word processing files that have been converted to **Portable Document Format** should not be included.

Accuracy of marking

The majority of centres submitted controlled assessment tasks to a very high standard and accurate marking was generally noted. A small number of centres marked leniently and marking was adjusted. Marking of candidate evaluations was an area of concern for some centres.

The majority of centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that **all** centres complete internal standardisation effectively before submitting marks to CCEA. Some centres did not follow the mark scheme and/or adhere to internal standardisation procedures.

It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshots or step by step reports for controlled assessment tasks. Thank you to all centres for their efforts this series with electronic controlled assessment submission.

Task 1 – Designing the multimedia and database solution using appropriate tools:

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- The majority of storyboards were computer generated. If 'hand-drawn' storyboards are used, they must be scanned to a high resolution to assist the moderation process. Rather than the focus on the quantity of storyboarding, candidates should focus on the quality and detail of storyboards, for example, the design of the website could

include the design of the masterpage. Database design may include a more text/ tabular form/query/report design, to reduce the duplication of work.

- Storyboards must show the sources of information and file types to be used.
- The design of professional reports is to be commended, with full titles and the company logo applied appropriately.
- However, many candidates failed to plan complex queries (two or more criteria) or appropriate reports (grouping, sorting and calculations).
- Some candidates did not attempt prototyping and the use of feedback in refining the solution, this section was marked leniently. 'Marking up' of prototype amendments and feedback could be applied to storyboards in a different font colour.

Task 2 – Building the multimedia and database solution:

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. The following points should be noted:

- Graphics, animation and videos produced for the candidate websites were generally of a high standard.
- Overall the standard of web authoring was high. It is positive to note many candidates using masterpages and cascading style sheets.
- Scripted elements may include: form validation, slideshow, animation, login, CSS, colour (hex).
- Accessibility elements may include: browser settings, plugins (downloads), speech, magnification, Alt tags.
- **The location of scripted and accessibility elements must be noted on the eCandidate record sheets.**
- Candidates should be encouraged to name their website homepage as 'index' or 'home'.
- Database solutions incorporated appropriate relationships, user friendly and intuitive switchboards, in addition to appropriate validation and input masks.
- Database solutions were implemented from the design document, however the depth of complex queries and reports (grouping, sorting and calculations) were lacking, particularly grouping and calculations.

Task 3 – Testing the database solution:

- Testing must include a range of valid, invalid and extreme data.
- Some candidates designed excessive tests and included screenshots for all testing cases, this could be further refined.
- No screenshots are required for testing; screenshots are only required to demonstrate corrective action.

Task 4 – Evaluating the multimedia and database solution:

- Some candidates described their solution, rather than develop evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met: fully met, partially met or not met.

- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness and refinements were not always given in depth consideration.
- This section was marked leniently by some centres.

Chief Examiner's Report

Assessment Unit 4 Digital Development Concepts

Overview

In this unit, candidates develop their understanding of programming concepts. This constitutes 40% of the GCSE Digital Technology (Programming). The paper this year was worth 120 marks and the time allocated for completion was 90 minutes. Questions focused on the skills candidates developed across the unit and on a number of computing concepts such as number systems, logic and algorithmic design.

Q1 Parts (a) and (b) of this question were well attempted by the majority of candidates who were able to identify the correct data types and the correct validation checks. Part (c) required candidates to complete a simple algorithm. Many candidates achieved top marks in this question for inputting the fourth score and calculating the average score. Candidates typically lost marks in Part (c) for (i) not outputting the average or inaccurate calculation of it. In Parts (e)(i) & (ii) most candidates demonstrated an understanding that the computer programs must be translated to binary because a computer cannot understand the code and articulated the role of the language compiler.

Part(d) was less well attempted but a significant number of candidates were able to name three elements associated with the Ide. Candidates were not awarded marks for answering 'colour', rather an explanation was required.

Q2 This question tested knowledge of variables and constants. A high proportion of candidates correctly answered Parts 2(a) and (b) showing a good understanding of these two aspects.

Part 2(c) gave candidates a scenario for which they were required to write a solution in algorithmic or code form. Either was acceptable. A significant number of candidates did not make use of a loop but instead repeated the input statements for the five hiking distances. Marks were allocated for the use of a loop in this situation. Marks for calculations were allocated if the calculation was correct; for example, a running total e.g. total = total + miles or total = miles1 + miles2 + miles3 + miles4 + miles5 were allocated the same marks. Generally, candidates in the top band achieved the marks for using a loop.

Candidates should be encouraged to use a loop for validation together with an appropriate Boolean variable or condition to control the loop.

Q3 A closed question which was well attempted, many candidates achieved full marks in this question showing a good understanding of conditions and logic.

Q4 A closed question which was well attempted, and again a high proportion of candidates achieved full marks in this question showing a good understanding of program structures. Some candidates answered selection for iteration and vice versa.

Q5 This question was about syntax and logic errors. Candidates were able to describe each type of error. Typical answers included 'program does not run with syntax errors' or 'logic errors occur when the program is running'. However, they were less well able to give an example of each. Many candidates were able to say that a syntax error may involve a spelling mistake but explaining a logic error or giving an example proved much more challenging. Top band candidates provided responses such as 'causes incorrect output' or 'causes unexpected behavior'. A significant number of candidates achieved 3 marks in this question because they were unable to provide

a tangible example of a logic error. This could be remedied by providing a sample program with logic errors for candidates to review and 'fix' during a practical session.

Q6 This question assessed QWC and related to computational thinking in terms of abstraction and decomposition. Candidates answers were across all bands. Most were able to define the two concepts, however only a minority were able to aptly give examples to illustrate or explain their answers. Candidates should be aware when developing solutions that they are in fact deploying some of the key elements of computational thinking this will help develop a deeper understanding of the approaches to problem solving going forward to Advanced level study.

Q7 Most candidates were able to expand the acronym ASCII accurately and a significant majority gained full marks in the closed response question in Part (b). Where candidates lost marks it was generally in the question parts which referenced Unicode. It is important that candidates are aware of the reasons for the development of Unicode.

Q8 This question was about number systems and candidates responses to Parts (a) and (b) were very good. A minority of candidates were unable to convert from binary to hexadecimal. In Part (c) 'colour' on its own was not acceptable, candidates must say at least 'colour codes'. The mark scheme was broadened to accept a range of answers.

Part (d) related to overflow and candidates achieved a mark for indicating that the overflow existed and a mark for showing carried values. It is worth pointing this out to candidates going forward so that their calculations are properly formulated so that marks can be allocated.

Candidates had some difficulty articulating a response to Part 8(d)(iii) How can overflow affect the result of a calculation? Many candidates were able to state that the answer would be incorrect but failed to mention that the overflow bit is lost, thereby only achieving 1 mark.

Candidates did well in completing Part (e) the truth table, many achieved full marks.

Q9 Overall this question was well attempted. Candidates were asked to call upon their knowledge of testing and use of data types.

(a) (i) In this part many candidates achieved only 1 mark as they were not able to give the range of tests required for robustness. Many stated valid and invalid but together these would only attract 1 mark of credit. Candidates should know that there are a variety of tests for robustness including the volume of data and the data values used (so that all pathways are tested). Testing types such as white box and black box testing were not accepted here.

(b) (i) Required candidates to write some validation code for use in a program. It should be noted that candidates should make use of a loop during validation as re-entry of data is part of that process. Almost all candidates included the error message to be output and attempted the condition relating to the number of frames, which a significant number got correct. Marks were lost where candidates did not use a loop and did not set a valid variable or condition to control the loop.

(ii) Showed a test plan and candidates were required to complete the test plan. This was well attempted with the exception of 'Null Data'. Candidates found it challenging to provide a relevant example of Null Data. Answers such as 'nothing' or '0' were allocated no marks. Candidates providing acceptable examples were in the minority.

(c) Worked well as a closed question and the majority of candidates across all bands were able to score in this question. Common mistakes included candidates using >20 rather than >=20 for the if statement; the former answer was allocated 1 mark only. The responses demonstrated that with scaffolding the majority of candidates can attempt an algorithmic solution.

Q10 This question focused on testing and examined candidates understanding of system testing, definitions of various methods of testing and the development of appropriate test plans.

Most candidates were correctly able to provide definitions of both black box and white box testing. For many candidates the responses contained reference to the fact that in black box testing the design or structure of the underlying solution was unknown by the tester whilst in white box testing the tester was familiar with the code.

Other common responses included reference to the fact that white box testing was carried out by a programmer or someone who have detailed knowledge of the code. In future examination of this topic area they examining team would expect to see an increase in the technical responses provided by candidates. For example: -

Black box – focuses on inputs/outputs, tests functionality, relates to how the interface operates.

White box – employs the user of trace tables, tested using a dry run process.

(b) Many candidates were able to select the correct statements about system and integration testing obtaining two marks.

Q11 The focus of this question was on the process of evaluation and use of a range of document which could support that process. Candidates referred more the the documents and their content rather than how the document could contribute to the evaluation process.

(a) While many candidates were able to reference the fact that the main aim of the evaluation process was to gain feedback from the end user, very few were awarded the second mark for this question. Schools should ensure candidates are aware of the need to conduct an evaluation to ensure that all user requirements have been met or to help improve the product being developed

(b) It was pleasing to see the majority of candidates were able to correctly reference two individuals involved in the evaluation process; reference to the developer/ programmer and the end user were the most common responses provided

(c) This question was poorly attempted. While candidates are familiar with the need and content of each of the document types referenced in each part of this question it is important also that they are aware how various document types are used to support the evaluation process.

(i) Only a small number of responses provided made reference to the use of the user requirements document as a check list or point of comparison; too many candidates instead referenced the content of the document; describing what the document contained rather than how it was used.

(ii) This question part was the least well answered part of this question. Common responses from candidates focused on the testing process rather than how test plans were used during evaluation, for example, to see if…., to make sure everything works. Only a small percentage of candidates were awarded full marks to this question part.

(iii) The focus of most answers in this question part referenced how the

design document contained plans for the proposed solution, rather than referencing the fact that these plans were used as a point of comparison either at the design stage or at the end of development.

Q12 This question about sorts and searching techniques allowed candidates to show their understanding through a practical sorting example.

(a) It is clear that most candidates understood the difference between the binary and linear search as the majority of candidates scored 1 mark here. However, in Part 12(b) candidates were less able to construct a response related to the number of attempts required to find a value for each of the searches. Many candidates achieved only 1 mark here as they did not refer to the reduction in the search list each after each attempt.

(c) Was a practical application of the bubble sort. Candidates were required to complete the sort order after each pass. Many candidates were unable to accurately do this in passes 1, 2 and 3. However, a significant number were able to get the sort order correct in pass 4. Candidates should attempt this process on paper and via a computer program to ensure they have a full understanding of the difference between comparison between adjacent elements and passes through the array. This is vital to instill an understanding of the efficiency of a sort.

Principal Moderator's Report

Assessment Unit 5 Digital Development Practice

Overview

All centres successfully submitted controlled assessment tasks electronically. Centres **must** ensure candidates have been provided with the correct scenario for Unit 5. This information is clearly provided on the front cover of the controlled assessment units. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised. The quality and assessment of candidate work has remained accurate.

Administration

The following procedures **must** be adhered to:

- Samples should be submitted on **memory pen (preferable)**, or **CD**, rather than DVD.
- Unit 5 samples must be submitted **on separate media**.
- **The contents of each sample must be verified** after the sample has been copied to media.
- All candidates of the requested sample, including the **top** and **(complete) bottom candidate** of the sample must be included. A number of centres did not include the top and bottom candidate this series.
- **Folder labels 'top'** and **'bottom'** must be included after the 'candidate number' label on the sample folder, for example, 'Candidate 0000 (Top)'.
- Only **essential files** should be submitted for moderation. Original data files or duplicate word processing files that have been converted to **Portable Document Format** should not be included. A small number of centres did not meet this particular requirement.

Accuracy of marking

The majority of centres submitted controlled assessment tasks to a very high standard and accurate marking was generally noted. A small number of centres marked leniently and marking was adjusted. Marking of candidate evaluations was an area of concern for some centres.

The majority of centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that **all** centres complete internal standardisation effectively before submitting marks to CCEA. Some centres did not follow the mark scheme and/or adhere to internal standardisation procedures.

It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshots or step by step reports for controlled assessment tasks. Thank you to all centres for their efforts this series with electronic controlled assessment submission.

Task 1 – Design a solution using appropriate tools:

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- In some instances, user requirements were extracted from the task, without development or interpretation. Target audience needs must also be fully identified.

- The design component was particularly well implemented with high quality flowcharts and pseudo code included effectively.
- Validation and error trapping techniques were designed appropriately, however use of user feedback to refine the solution was omitted by a number of centres and marked leniently.

Task 2 – Building a solution:

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. Centres are to be highly commended upon their approach to this task. The following points should be noted:

- **Program usernames and passwords must be noted on the eCandidate record sheets.**
- Marking was highly accurate across the sampled centres and the programming solutions were of a high standard.
- Solutions of the highest standard gave due consideration to system robustness.
- All solutions must be annotated.

Task 3 – Testing the solution:

- Generally, test plans were well structured, in tabular format (showing tests, expected outcome, actual outcome and fixes), in some cases test plans were cluttered and could be more professionally presented.
- It is very important that testing incorporates a range of tests (black and white box testing) and includes valid, invalid and extreme test data.
- Some candidates designed excessive tests and included screenshots for all testing cases, this could be further refined.
- ***No screenshots are required for testing; screenshots are only required to demonstrate corrective action.***

Task 4 – Evaluating the solution:

- Some candidates described their solution, rather than develop evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met: fully met, partially met or not met.
- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness and refinements were not always given in depth consideration.
- This section was marked accurately by most centres, however a small number of centres marked performance, robustness and refinements section leniently.

Contact details

The following information provides contact details for key staff members:

- **Specification Support Officer: Nuala Tierney**
  (telephone: (028) 9026 1200, extension: 2292, email: ntierney@ccea.org.uk)
- **Officer with Subject Responsibility: Andrew Douglas**
  (telephone: (028) 9026 1200, extension: 2713, email: adouglas@ccea.org.uk)

© CCEA 2019

COUNCIL FOR THE **CURRICULUM, EXAMINATIONS** AND **ASSESSMENT**
29 Clarendon Road, Clarendon Dock, Belfast BT1 3BG
Tel: +44 (0)28 9026 1200 Fax: +44 (0)28 9026 1234
Email: info@ccea.org.uk Web: www.ccea.org.uk

Rewarding Learning